The irreductionist hermeneutics of the Grounded AI Map

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The *Grounded AI Map* is a giant "walkable visualization" (Rodighiero, 2018) of how AI and algorithms are involved in science, designed by analyzing 2 million scientific papers retrieved from Scopus (Fig. 1 & 2). The map consists of a 100 square meters floor mat that participants can explore physically by walking on its surface and interact through a smartphone app. It shows one dot for each paper and four thousand labels of thematic clusters. The visualization has been exhibited in multiple public venues in 2025, including the Danish Technical Museum (Fig. 3), and has been adapted for audiences in both Danish and English. The mobile website includes three modalities of information: text, audio, and visual data.

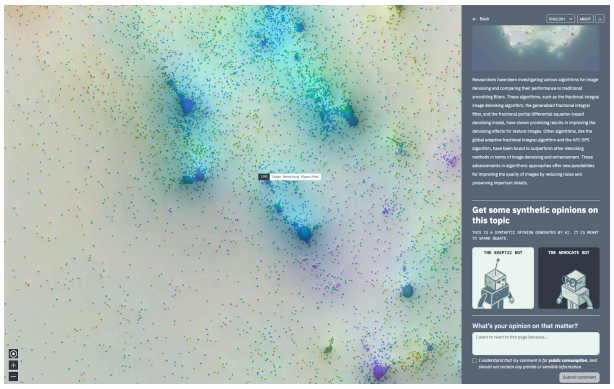

Figure 1: The *Grounded AI Map* visualizes 2 million scientific articles extracted from Scopus.

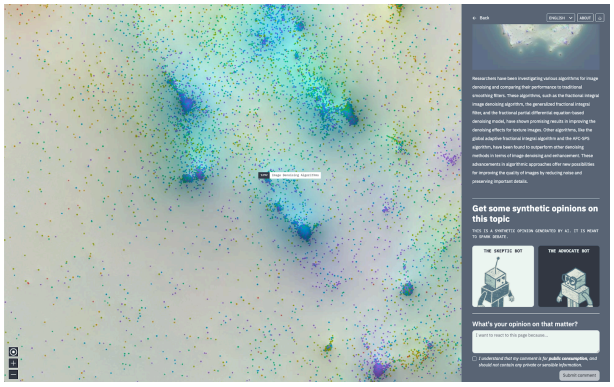

Figure 2: Close-up of the map: each dot is a scientific article, annotations indicate specialties.

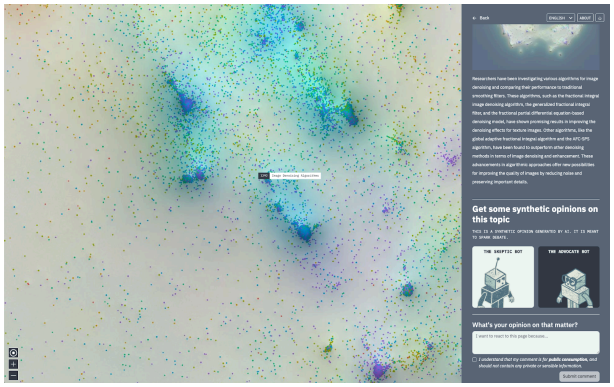

Figure 3: Exhibition at the Danish Technical Museum.

Exhibition at DH2025

We propose to exhibit the installation at the DH conference, producing a smaller scale version if necessary. Our primary intention is to engage the public by providing a first-hand aesthetic experience featuring a distant reading that allows the visitors to move throughout different scales of abstraction of the 2 million scientific papers mapped.

Method

We harvested a corpus of approximately two million scientific articles from the *Scopus* database, using the following query in the title or abstract: *algorithm*, AI, "artificial intelligence", or "machine learning"*. The query is deliberately limited, yielding an incomplete yet broad representation of the AI discourse in science.

We computed connections between articles using text analysis and natural language processing techniques. We associated each abstract to a vector using an embedding model trained on scientific literature (Cohan et al., 2020). Then we extracted a network of semantic similarity from the vectors using the k-nearest neighbors approach, where nodes are documents and edges are semantic similarities.

We computed a placement of the documents using a force-driven layout algorithm (Jacomy et al., 2014), allowing clusters of similar documents to emerge. We rendered the map by only displaying the nodes, as dots, color indicating the publication year, from red (old), to yellow, and to blue (recent).

We extracted the densest clusters in the embedding space to identify over 4000 scientific specialties. Using a large language model and a specific prompt, we summarized how AI and algorithms are involved in each specialty, and generated a title. We validated this method qualitatively and quantitatively. We used the titles to annotate the map and make it readable without aggregating the data points, avoiding a reductionist approach.

Design philosophy

We focused on a hermeneutic engagement with the corpus through different forms of interaction. Although our method involves computing semantic distances with an AI embedding and retrieving clusters of similar documents, the initial data is never reduced to those clusters but instead annotated with them to minimize the loss of information.

We call this approach “irreductionist”, following the notion that phenomena and documents are neither reducible nor irreducible in essence (Latour, 1984). We do not reject all reductions (and our map uses some), but we reject more broadly that meaning is necessarily based on reduction. We conceived our map as a mediator between a corpus of 2 million papers involving AI and algorithms, and the viewers, specifically aimed to provide a reversible translation by making use of reduced affordances (annotations) and the non-reduced source (the dots).

We argue that despite the reduction of two millions documents into four thousands annotations, the presence of the dots themselves allows engaging with features that would otherwise get lost in reduction. For instance, the clusters aren’t entirely separate entities as suggested by the algorithm; they often overlap, stretch, collide, and generate numerous unique patterns that can be directly observed on the map. We attempted to implement “graphical means to produce interpretative work using visual argument structures such as contradiction, ambiguity, parallax, and point of view that are fundamentally hermeneutic in character” (Drucker, 2018).

We accompanied the map with a website and smartphone app offering support for engaging in dialogue with it (Fig. 4). It features the “slanted bots”, AI agents with distinct personalities: one optimistic and progress-oriented, the other sceptical and critical of AI. Commenting on each cluster when navigating the app, they contribute to keeping the meaning of the map open to interpretation.

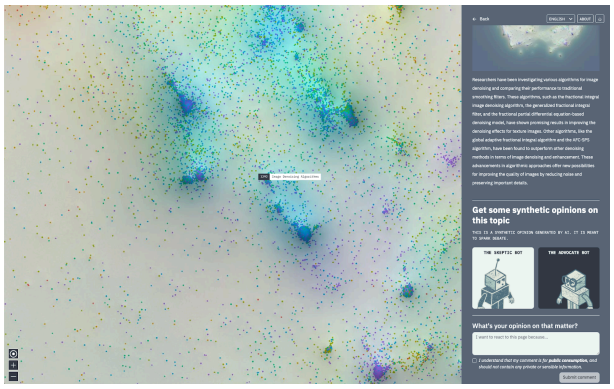

Figure 4: Online app allowing to navigate the map and interact with the “slanted bots”.

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